

DATA PIPELINE_01: BLOB to SQLDB (ADF USING DATAFLOW)

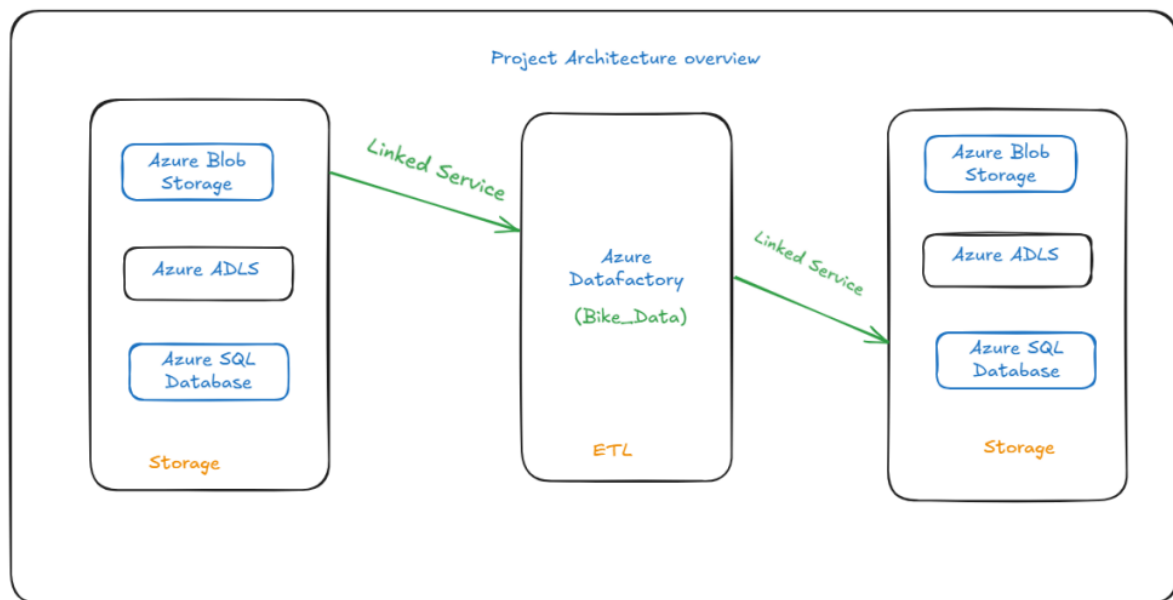
Project Objective:

To extract raw data which is stored in Azure Blob Storage, transform it using Data Flow in Azure Data Factory, and load it into Azure SQL Database.

Process:

Copy File from Blob Storage → Transform (ADF) → Load into SQL Database (via ADF)

Architecture:



What Is BLOB:

Azure Blob Storage is Microsoft's cloud-based object storage solution designed to store massive amounts of unstructured data. It is ideal for files that don't fit into traditional databases, such as images, videos, audio, backups, and log files.

Step 1: Sign in to Azure Portal Go to <https://portal.azure.com> and Sign in with your Microsoft/Azure account

Step 2: 3. Create an Azure Storage Account ---Blob storage called a storage account — like a big folder in the cloud.

Azure services



- Click "Create a resource"

Home

Create a resource



I need a new low-cost VM

Get Started

Recently created

Categories

Machine Learning

AI Apps and Agents

Analytics

Blockchain

Infrastructure Services

Databases

Developer Tools

DevOps

Identity

Integration

Internet of Things

IT & Management Tools

storage



Getting started? Try our Quickstart Center

Create



Web App

Create



Key Vault

Create



Virtual machine

Create



Storage account

Create



Logic App

Create | Learn more



Windows 11 Enterprise, version 25H2

Create | Learn more



Ubuntu Pro 24.04 LTS

Create | Learn more



Free SQL Server License: SQL Server 2022 Developer on ...

Create | Learn more



Red Hat Enterprise Linux10 (latest minor version)

Create | Learn more



Debian 13 "Trixie"

- Search for "Storage account" and click Create

Fill basic details:

Subscription: Choose your subscription: Azure Subscription1

Resource Group: Select or create a new one : nk-rg

Storage account name: Unique name (lowercase only) : nkblobsa

Region: Pick a location close to your users : Asia specific Southindia

Primary service : Azure Blob or Azure Data Lake Storage Gen2

Performance: Standard (default is fine)

Redundancy: Choose Locally-redundant storage (LRS) for basic use.

Fill Advanced details:

Access Tier: Hot, Cool, or Cold based on usage.

Fill Networking Details:

Network Access: Enable Public access from all networks (for practice).

Routing Preference: microsoft Network Routing.

Fill Data Protection details:

Soft delete for blobs: Enable (recommended)

Soft delete for containers: Optional.

Soft delete for file shares: Optional

Click Review + Create, then Create

Home > Create a resource

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription	NK
Resource group	NK-RG
Location	South India
Storage account name	nkblobsa
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

[Previous](#) [Next](#) [Create](#)

*** Initializing deployment...
Initializing template deployment to resource group.

Sucessfully created blob storage

Home

nkblobsa_1782196969836 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name: nkblobsa_1782196969836
Subscription: [NK](#)
Resource group: [NK-RG](#)

Start time: 23/6/2026, 12:14:19 pm
Correlation ID: e2223130-d522-4815-9f82-6354d4cff0d3

Deployment details

Next steps

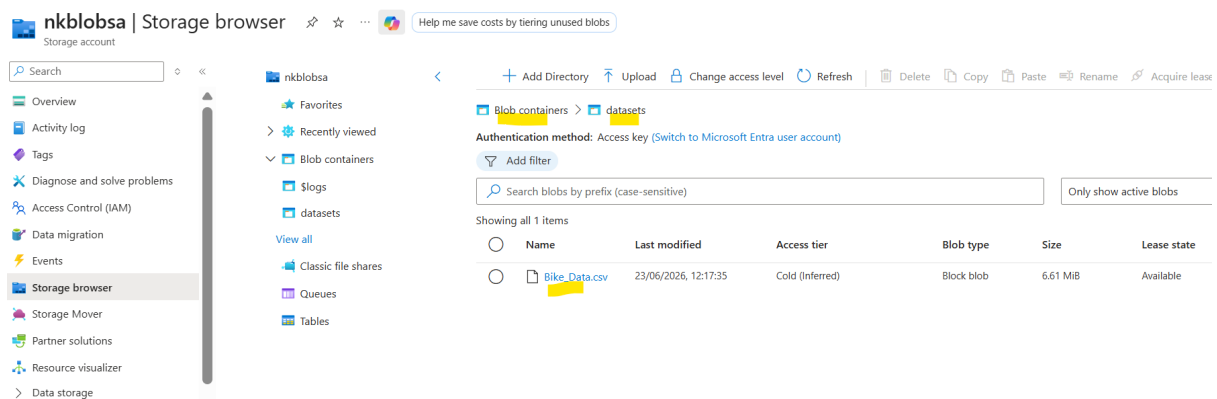
[Go to resource](#)

Give feedback
[Tell us about your experience with deployment](#)

Create a Blob Container:

Containers are like folders inside the storage account to organize files.

- Click "+ Container"
- Give it a name: datasets
- Click Create
- Upload the raw data



Create a SQL Database:

- Click "Create a resource" (top left)
- Search for "SQL Database", then click Create
- Fill in the basics:
 - o Subscription: Choose your subscription

Resource Group: Select the one you created

Database name: Give database name

Server: Click "Create new"

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Resource group * ⓘ

[Create new](#)

Create a SQL Server

- This is where your database will present.
- Click Create new (under SQL Server)

Server name: (must be unique)

Location: Same as resource group

Admin login: Create a username

Password: Set a strong password o Click OK to return to the SQL DB creation form.

Configure Database:

Choose Basic pricing (DTU Basic)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server * ⓘ 

[Create new](#)

Enable Public Access (Firewall Rule):

Connectivity Method: Public Endpoint

In the Networking tab, under Firewall rules, select: Allow Azure services and resources to access this server: yes

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#) 

Connectivity method * ⓘ

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#) 

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

Add current client IP address *

- Click Review + create
- Wait for validation
- Click Create

Home

Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview

Search

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_d5bd... Start time : 23/6/2026, 12:38:35 pm
Subscription : NK Correlation ID : 2ed2ff7e-3ef5-4722-b0c5-8e2d61ef9170
Resource group : NK-RG

Deployment details

Resource	Type	Status	Operation details
nkserver	Microsoft.Sql/servers	Accepted	Operation details

Sucessfully created the sql database

Home

Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview

Search

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_d5bd... Start time : 23/6/2026, 12:38:45 pm
Subscription : NK Correlation ID : 2ed2ff7e-3ef5-4722-b0c5-8e2d61ef9170
Resource group : NK-RG

Deployment details

Next steps

[Go to resource](#)

Go to Resource—Query editor(preview)---Provide the Sql authentication details(username and password)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_d5bde4d31f7d402d86326 | Overview > free-sql-db-4012917 (nkserver/free-sql-db-4012917)

free-sql-db-4012917 (nkserver/free-sql-db-4012917) | Query editor (preview)

Search

Feedback

Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Add or remove favorites by pressing Ctrl+Shift+F

Run queries on your database directly in the browser [Learn more](#)

Connecting to free-sql-db-4012917 on nkserver

Microsoft Entra authentication

SQL authentication

Username *

Enter username

Password *

Enter password

Connect

[Forgot password? Reset in logical server settings](#)

**free-sql-db-4012917** (nkserver/free-sql-db-4012917) | Query editor (preview) ☆ ...
SQL database

◇ << >>  New query  Templates ▾  Open in ▾  Feedback  Sign out  Classic experience

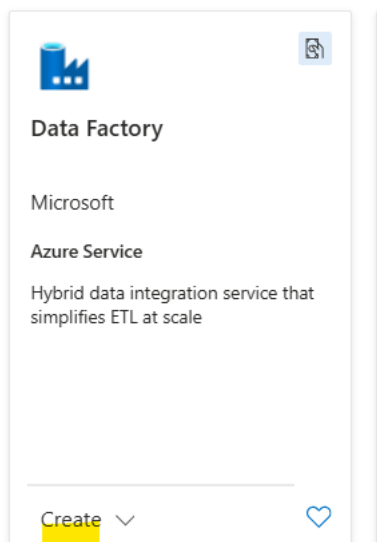
 Overview
 Activity log
 Tags
 Diagnose and solve problems
 **Query editor (preview)**
 Mirror database in Fabric (preview)
 Resource visualizer
> Settings
> Data management
> Integrations

Explorer <<
 ≡
▽ free-sql-db-4012917
 >  dbo
 ▽  Queries


Start exploring
Write and run SQL queries, create database objects
preview your

Create Azure Data Factory :

- Search for Data Factory > Click Create



Fill in the basic details required

Basics

Subscription

Resource group: NK-RG

Name: nk-azuredataf

Region: South India

Version: V2

Networking:

Connect via:Public endpoint

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

NK

Resource group * ⓘ

NK-RG

Create new

Instance details

Name * ⓘ

nk-azuredatf

Region * ⓘ

(Asia Pacific) South India

Version * ⓘ

V2

Home > Create a resource > Marketplace

Create Data Factory ...

- Basics
- Git configuration
- Networking
- Advanced
- Tags
- Review + create

[View automation template](#)

Basics

Subscription	NK
Resource group	NK-RG
Name	nk-azuredatf
Region	South India
Version	V2

Networking

Connect via	Public endpoint
-------------	-----------------

Home



Microsoft.DataFactory-20260623124515 | Overview ...

Deployment

Search



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : Microsoft.DataFactory-20260623124515

Subscription : NK

Resource group : NK-RG

Start time : 23/6/202

Correlation ID : 364d24b



Deployment details



Next steps

[Go to resource](#)

Successfully created azure data factory

Resources

Recommendations



Filter for any field...

Type equals all

Location equals all

+ Add filter



Name ↑

Type

Location



free-sql-db-4012917 (nkserver/free-sql-db-4012917)



SQL database

South India



nk-azuredat



Data factory (V2)

South India



nkblobsa



Storage account

South India



nkserver



SQL server

South India



Delete

Resource group (move) : [NK-RG](#)

Type : Data factory (V2)

Status : Succeeded

Getting started : [Quick start](#)

Location : South India

Subscription (move) : [NK](#)

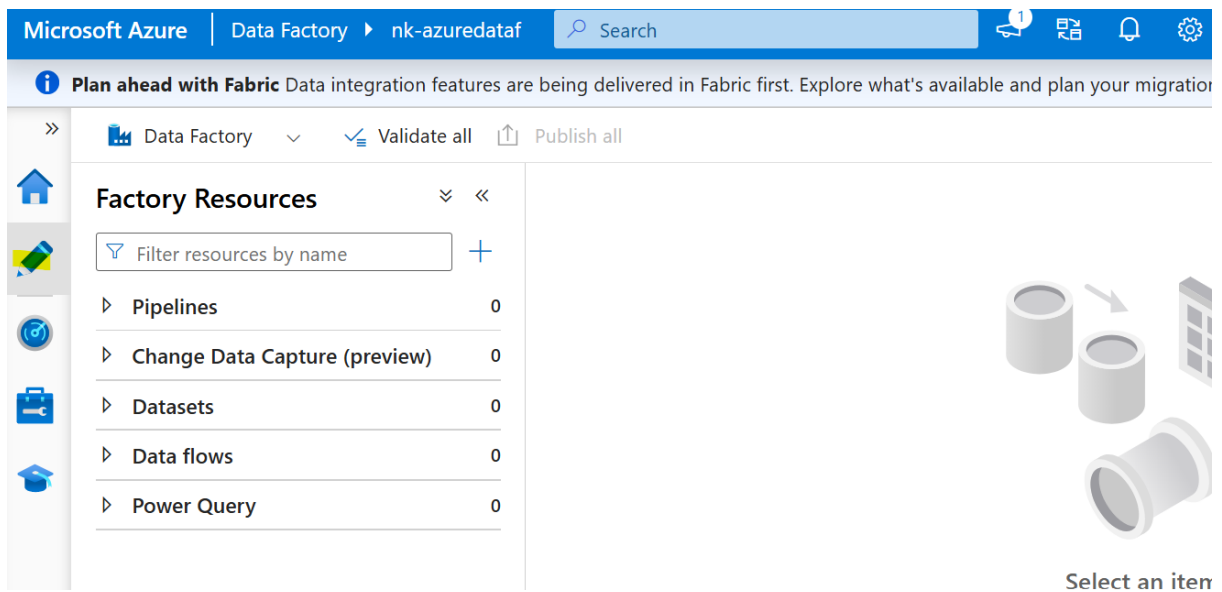
Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f



Azure Data Factory Studio

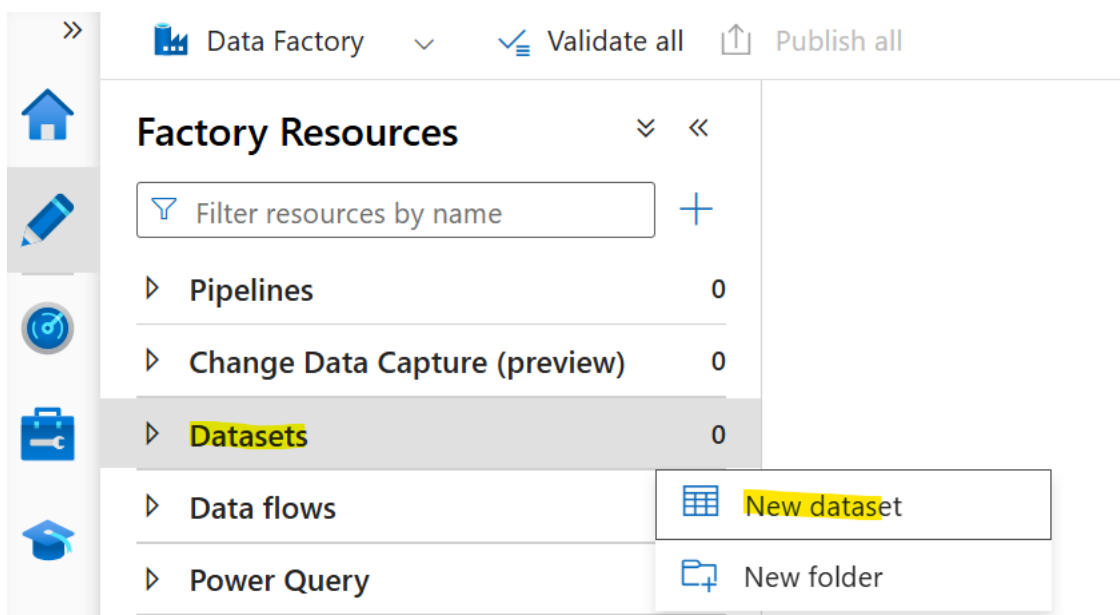
[Launch studio](#)

ADF Studio:



Step 1 in ADF:

We need to create a copy of the raw data from the storage account



Click on the new data set and select from which storage are we taking the copy of the raw data

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#) 

Select a data store

 Search

<

All

Azure

Database

File

Generic protocol

>


Azure AI Search


Azure Blob Storage


Azure Cosmos DB for
MongoDB

Continue

Cancel

Select the format of the data and click on continue

Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
	 JSON	

Continue

Back

Cancel

A new popup displays give the name and add the linked service to push the copy data from storage

Set properties

Name

InputData

Linked service *

Select...

First row as header



OK

Back

Cancel

Create the linked service :

1. Go to your ADF → Click Launch Studio 2. In left menu → Click Manage (gear icon)
3. Under Linked Services → Click + New
2. For Azure Blob Storage • Choose Azure Blob Storage • Auth type: Account key or Azure AD • Select your storage account • Click Create

Set properties

Name

InputData

Linked service *

blob2adf

File path

datasets / Directory / Bike_Data.csv

First row as header

☒

Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Plan ahead with Fabric Data integration features are

>>

Data Factory

Validate all

Factory Resources

Filter resources by name

Pipelines0

Change Data Capture (preview)0

Datasets1

InputData

Data flows0

Power Query0

Click on publish all to save the changes in adf

Plan ahead with Fabric

Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Data Factory

Validate all

Publish all 1

Factory Resources

Filter resources by name

Pipelines0

Change Data Capture (preview)0

Datasets1

InputData

Data flows0

Power Query0

InputData

DelimitedText

InputData

Connection

Schema

Parameters

Linked service *blob2adf

Test connection Edit + New [Learn more](#)

File path *datasets / Directory

Compression typeNo compression

Column delimiter ①Comma (,)

Row delimiter ①Default (\r\n, or \r\n)

Encoding ①Default(UTF-8)

We can also preview the data

Connection

Schema

Parameters

blob2adf

ew [Learn more](#)

tasets / Directory / Bike_Data.csv

Browse

Preview data

compression

mma (,)

efault (\r\n, or \r\n)

efault(UTF-8)

Preview data

Linked service: blob2adf

Object: Bike_Data.csv

	Region	Country	Customer	BusinessSegment	Category	Model	Color	SalesDate	ListPrice
1	North America	United States	Professional Containers and Packaging Co.	Clothing	Tights	Women's Tights	Black	22-08-2018	74.99
2	North America	United States	Professional Containers and Packaging Co.	Clothing	Gloves	Full-Finger Gloves	Black	17-08-2018	37.99
3	North America	United States	Professional Containers and Packaging Co.	Bikes	Mountain Bikes	Mountain-200	Black	14-08-2018	2294.9
4	North	United	Professional Containers and	Clothing	Jerseys	Long-Sleeve	Multi	13-08-	49.99

Create Dataflow for Transformations:

Plan ahead with Fabric Data integration features are being delivered in Fabric first

>> Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
- Data flows** 0
- Power Query

InputData

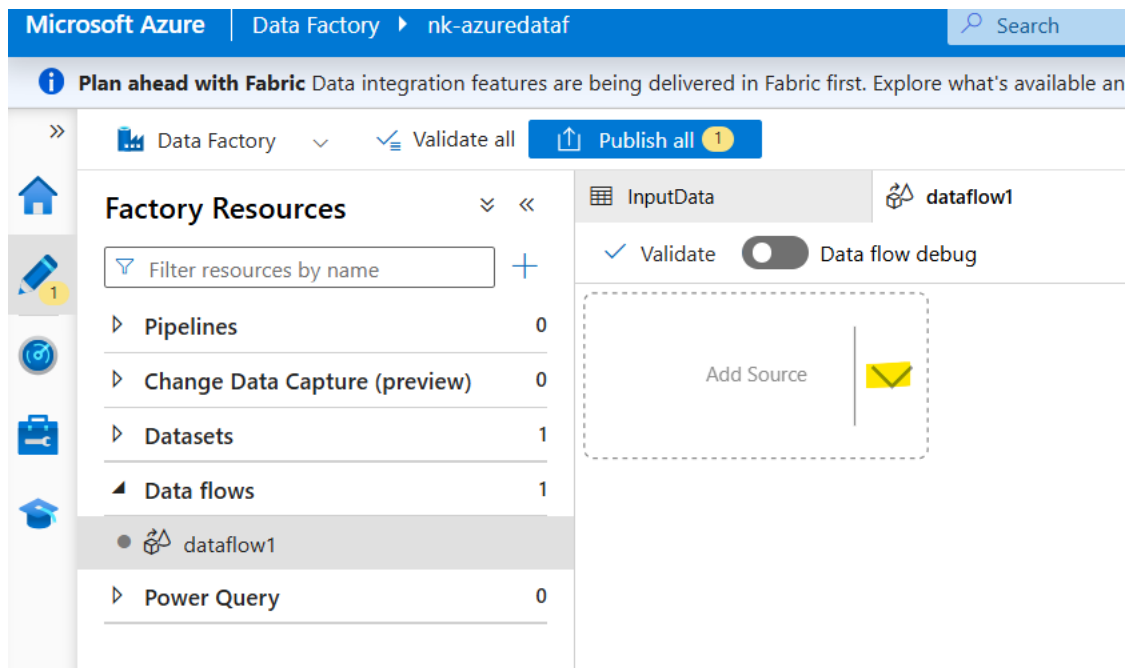
Delimited InputData

CSV

New data flow

New flowlet

New folder



Data Flow: Transformations

Source setting:

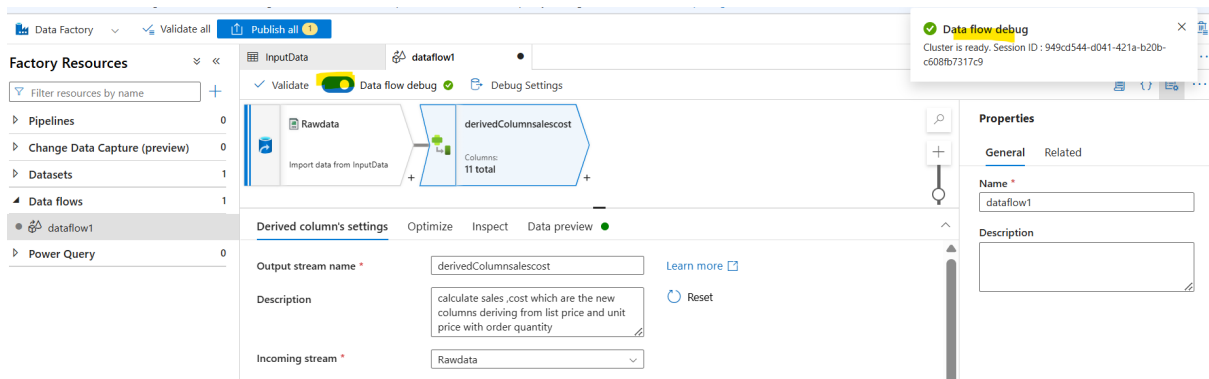
Select the rawdata that we brought from blob

- Use first row as header (if CSV)
- Inspect schema

Output stream name: can give any name for source

Source settings	Source options	Projection	Optimize	Inspect	Data preview
Output stream name *	Rawdata		Learn more		
Description	Import data from <u>InputData</u>		Reset		
Source type *	Dataset Inline				
Dataset *	InputData		Test connection Open New		
Options	☒ Allow schema drift ⓘ ☐ Infer drifted column types ⓘ				

Enable Dataflow Debug mode for 1 hr: (To see the transforamtion then & there)

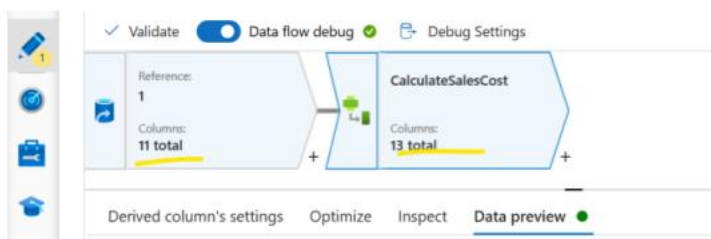


Derived Column (Add 3 new columns) :

Click + Add → Derived Column

Add 2 columns: Sales = ListPrice * OrderQty

Cost = UnitPrice * OrderQty



Note:

*****We cannot add the derived column on top of other derived column in this case used other derived column and add it

Profit = Sales - Cost Tip: You can use expression builder.

*****if there is any data mismatch then go to the source in dataflow----projection----change the datatype as required

Factory Resources

Filter resources by name

Pipelines 0

Change Data Capture (preview) 0

Datasets 1

InputData

Data flows 1

Bikedata_df

Power Query 0

InputData

Rawdata

Import data from InputData

derivedColumnSalescost

Creating/updating the columns: Region, Country, Customer, BusinessSegment, Category, Model, Color, SalesDate

derivedColumnProfit

Columns: 14 total

Derived column's settings

Optimize

Inspect

Data preview

Number of rows: INSERT 100, UPDATE 0, DELETE 0, UPSERT 0

Refresh, Typecast, Modify, Map drifted, Statistics, Remove, Export to CSV

Region	Country	Customer	BusinessSegment	Category	Model	Color	SalesDate
North America	United States	Professional Containers and P...	Clothing	Tights	Women's Tights	Black	22-08-2018
North America	United States	Professional Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	17-08-2018
North America	United States	Professional Containers and P...	Bikes	Mountain Bikes	Mountain-200	Black	14-08-2018
North America	United States	Professional Containers and P...	Clothing	Jerseys	Long-Sleeve Logo Jersey	Multi	13-08-2018
North America	United States	Professional Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	09-08-2018
North America	United States	Professional Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	06-08-2018
North America	United States	Professional Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	06-08-2018

Sucessfully derived 3 columns

Derived column's settings

Optimize

Inspect

Data preview

DELETE 0, UPSERT 0, LOOKUP 0, ERROR 0, TOTAL 1000

Export to CSV

Category	Model	Color	SalesDate	ListPrice	UnitPrice	OrderQty	cost	sales	Profit
Tights	Women's Tights	Black	22-08-2018	74.99	44.99	1	44.99	74.99	29.999999999999993
Gloves	Full-Finger Gloves	Black	17-08-2018	37.99	22.79	2	45.58	75.98	30.400000000000006
Mountain Bikes	Mountain-200	Black	14-08-2018	2294.99	1229.46	1	1229.46	2294.99	1065.5299999999997
Jerseys	Long-Sleeve Logo Jersey	Multi	13-08-2018	49.99	28.84	6	173.04	299.94	126.9
Gloves	Full-Finger Gloves	Black	09-08-2018	37.99	22.79	4	91.16	151.96	60.800000000000001
Gloves	Full-Finger Gloves	Black	06-08-2018	37.99	22.79	6	136.74	227.94	91.199999999999999
Canoe	Curious Can	Multi	06-08-2018	8.99	5.19	2	10.38	17.98	7.6

Select Columns:

- Removing unwanted columns

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

Filter resources by name

Pipelines 0

Change Data Capture (preview) 0

Datasets 1

InputData

Data flows 1

Bikedata_df

Power Query 0

InputData

Rawdata

Import data from InputData

derivedColumnSalescost

Creating/updating the columns: Region, Country, Customer, BusinessSegment, Category, Model, Color, SalesDate

derivedColumnProfit

deriving a new column profit

selected.columns

Columns: 12 total

Select settings

Optimize

Inspect

Data preview

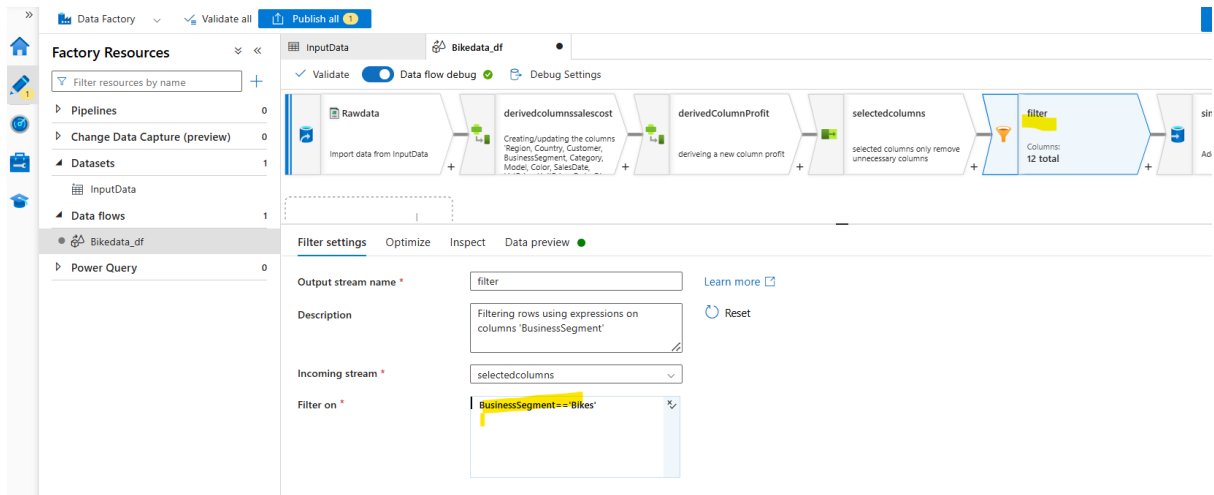
UPDATE 0, DELETE 0, UPSERT 0, LOOKUP 0, ERROR 0, TOTAL 1000

Map drifted, Statistics, Remove, Export to CSV

BusinessSegment	Category	Model	Color	SalesDate	OrderQty	cost	sales	Profit	
Containers and P...	Clothing	Tights	Women's Tights	Black	22-08-2018	1	44.99	74.99	29.999999999999993
Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	17-08-2018	2	45.58	75.98	30.400000000000006
Containers and P...	Bikes	Mountain Bikes	Mountain-200	Black	14-08-2018	1	1229.46	2294.99	1065.5299999999997
Containers and P...	Clothing	Jerseys	Long-Sleeve Logo Jersey	Multi	13-08-2018	6	173.04	299.94	126.9
Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	09-08-2018	4	91.16	151.96	60.800000000000001
Containers and P...	Clothing	Gloves	Full-Finger Gloves	Black	06-08-2018	6	136.74	227.94	91.199999999999999
Containers and P...	Clothing	Canoe	Curious Can	Multi	06-08-2018	2	10.38	17.98	7.6

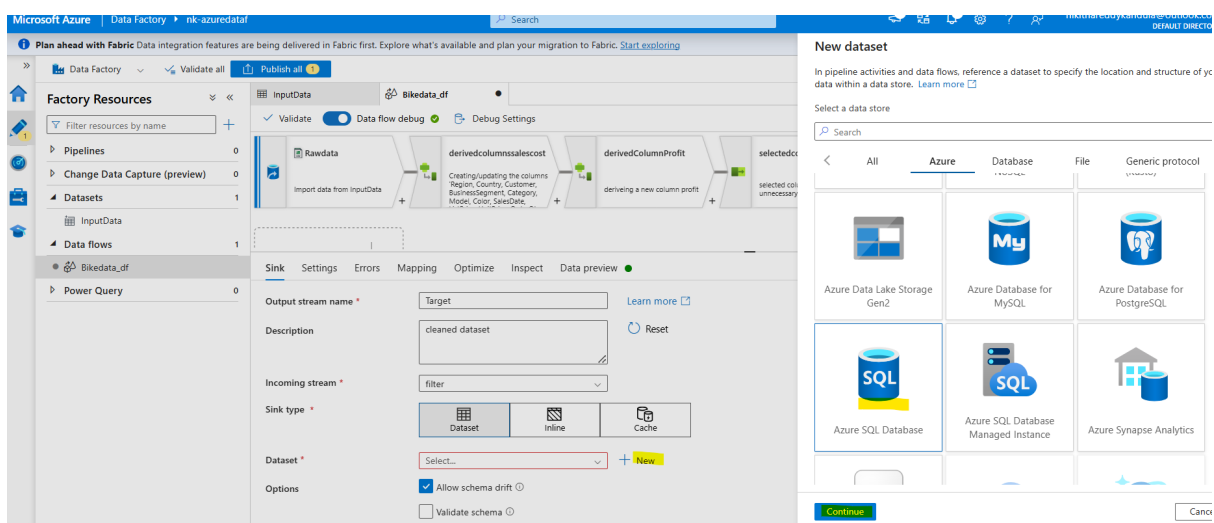
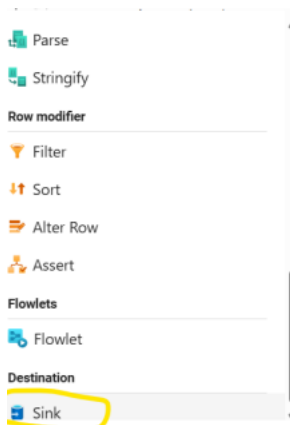
Filter Rows:

- Filtering only Bike Business Segment:
- Click + Add → Filter



Sink (SQL Table Output):

- Click + Add → Sink • Choose New dataset > Azure SQL Database • Select your linked service • Name: Sink_SQLDB • In sink settings: o Table name: e.g., dbo.EmployeeTransformed o Choose Auto create table



To Store Cleaned Data we need to create another Dataset: • Choose Azure SQL Database • Enter: o Server name, DB name o SQL Admin username and password • Test connection and click Create

New linked service

 Azure SQL Database [Learn more](#) 

Name *

AzureSqlDatabase1

Description

Connect via integration runtime * 

 AutoResolveIntegrationRuntime

Version

☒ 2.0 (Recommended) ☐ 1.0

 Import from connection string

Account selection method 

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Server name *

nkserver

Database name *

free-sql-dh-4012917

 Connection successful

Create

Cancel

 Test connection

Set properties

Name

adf2sqldb

Linked service *

AzureSqlDatabase1

☐ Select from existing table ☒ New table

Schema and table name

Schema name

OutputData

> Advanced

OK Back Cancel

>> Data Factory Validate all

Factory Resources

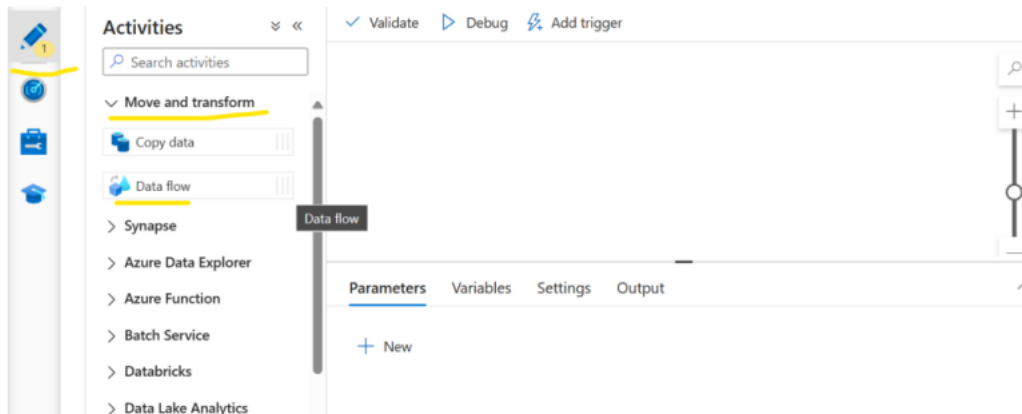
Filter resources by name

- Pipelines 0
- Change Data Capture (preview) ...
- Datasets 2
 - OutputData
 - InputData
- Data flows 1
 - Bikedata_df
- Power Query 0

Publish the changes in adf

Add Data Flow to Pipeline :

Go to Author → + New Pipeline o Drag Data Flow activity into canvas o Link it to the data flow you just created o Debug or Trigger the pipeline



Factory Resources

- Pipelines: 1
 - pipeline1
- Datasets: 2
 - OutputData
 - InputData
- Data flows: 1
 - Bikedata_df
- Power Query: 0

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Data flow settings:

- Data flow: Bikedata_df
- Run on (Azure IR): AutoResolveIntegrationRuntime
- Compute size: Small
- Logging level: Verbose

Activity runs

Pipeline run ID: b961b55e-bbad-43bc-a5b7-6476aaa7985b

All status

[Monitor in Azure Metrics](#) [Export to CSV](#)

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID	Log
Data flow1	Succeeded	Data flow	6/23/2026, 1:58:21 PM	2m 50s	AutoResolveIntegrationRuntime (South India)		ccf22b1c-19fd-4bb6-ad17-b03bfa9dea7d	